Implement SQL Operators

SQL operators are used to perform calculations, compare values, and combine conditions in SQL queries.

A. Arithmetic Operators

Arithmetic operators are used to perform mathematical calculations.

Operators: +, -, *, /, %

Example:

SELECT 10 + 5 AS Addition;

SELECT 10 - 5 AS Subtraction;

SELECT 10 * 5 AS Multiplication;

SELECT 10 / 5 AS Division;

SELECT 10 % 3 AS Modulus;

Output:

Addition	Subtraction	Multiplication	Division	Modulus
15	5	50	2	1

B. Logical Operators

Logical operators are used to combine multiple conditions.

Operators: AND, OR, NOT

Example:

SELECT * FROM Employee

WHERE salary > 20000 AND age > 20;

SELECT * FROM Employee

WHERE salary > 30000 OR age > 25;

SELECT * FROM Employee

WHERE NOT age > 25;

Explanation:

- AND – Both conditions must be true.
- OR – At least one condition must be true.
- NOT – Reverses the condition.

C. Comparison Operators

Comparison operators are used to compare two values.

Operators: =, >, <, >=, <=, <>

Example:

SELECT * FROM Employee WHERE salary = 25000;

SELECT * FROM Employee WHERE salary > 25000;

SELECT * FROM Employee WHERE salary < 25000;

SELECT * FROM Employee WHERE salary >= 25000;

SELECT * FROM Employee WHERE salary <= 25000;

SELECT * FROM Employee WHERE salary <> 25000;

Explanation:

Operator	Meaning
=	Equal to
>	Greater than
<	Less than
>=	Greater than or equal to
<=	Less than or equal to
<>	Not equal to

Implement SQL Operators

A. Special Operators

Special operators are used to check values against specific conditions.

1. BETWEEN Operator

Used to select values within a range.

```
SELECT * FROM Employee
WHERE salary BETWEEN 20000 AND 40000;
```

2. IN Operator

Used to match values from a given list.

```
SELECT * FROM Employee
WHERE age IN (20, 22, 25);
```

3. LIKE Operator

Used to search for a specific pattern.

```
SELECT * FROM Employee
WHERE name LIKE 'A%';
```

This displays employees whose names start with A.

4. IS NULL Operator

Used to check for NULL values.

```
SELECT * FROM Employee
WHERE salary IS NULL;
```

5. NOT BETWEEN Operator

Used to select values outside a specified range.

```
SELECT * FROM Employee
WHERE salary NOT BETWEEN 20000 AND 40000;
```

Special Operators Summary

Operator	Purpose
BETWEEN	Checks a range
IN	Matches values in a list
LIKE	Matches a pattern
IS NULL	Checks NULL values

Operator	Purpose
NOT BETWEEN	Checks values outside a range

B. Set Operation – UNION

The UNION operator combines the results of two or more SELECT statements and removes duplicate rows.

Example:

Assume two tables: Student1 and Student2.

```
SELECT name FROM Student1
UNION
SELECT name FROM Student2;
```

Example output:

Suppose the tables contain the following names.

Student1

name
Rahul
Priya

Student2

name
Priya
Amit

Output of UNION

name
Rahul
Priya
Amit

Explanation:

- UNION combines the results of both queries.
- Duplicate values are removed.
- Both SELECT statements must return the same number of columns, with compatible data types.

Note: UNION ALL combines the results without removing duplicates.